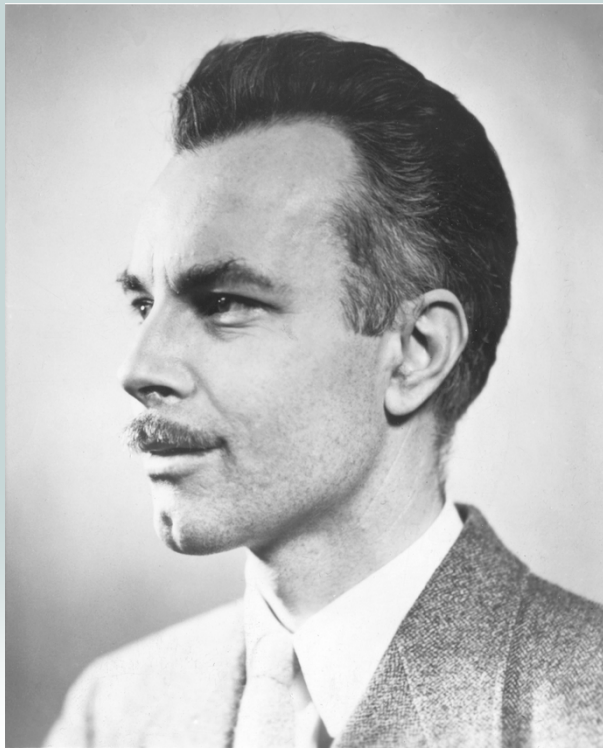




George Arthur Philbrick



Philbrick operational amplifier, circa 1952

Lecture 4: Operational Amplifiers (Op Amps)

Overview

The operational amplifier, or op amp for short, is a fundamental building block in circuit design. Stuffed inside a chip are a bunch of transistors and other elements that make up a near-ideal voltage controlled voltage source (VCVS) with near - infinite gain. This creates special conditions that simplify circuit analysis. The op amp is a basic building block for a number of circuit applications, and you'll learn how to choose the right circuit and how to specify the correct parameters to achieve a specified function - the first steps in design. Finally, you'll learn where the ideal op amp model breaks down, and how this affects circuit operation and design.

Key Term

•Gain

Ratio of the output to the input. An amplifier whose output voltage was 10 times the input voltage has a gain of 10.

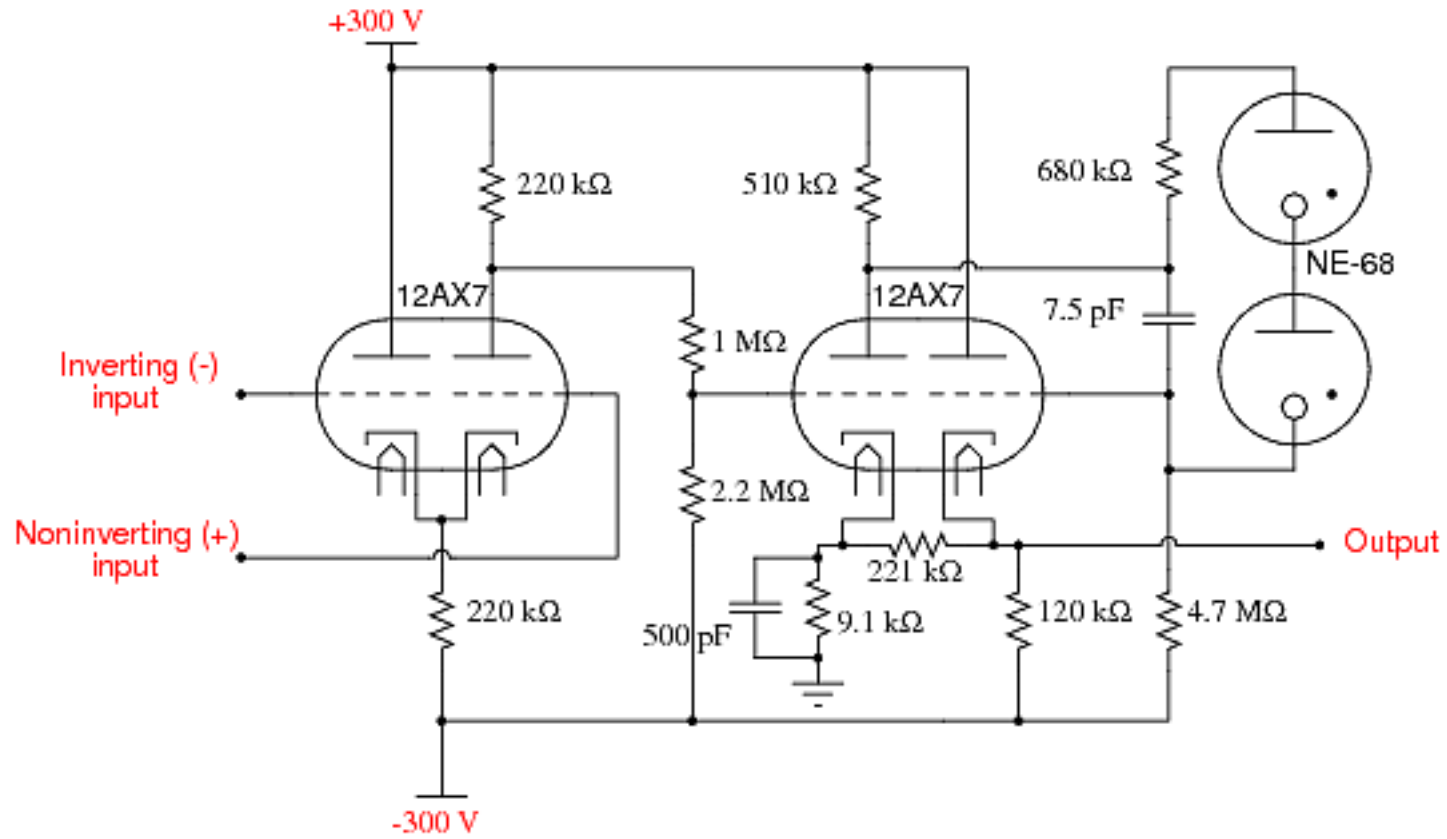
Learning Objectives

At the end of this lesson, you will be able to

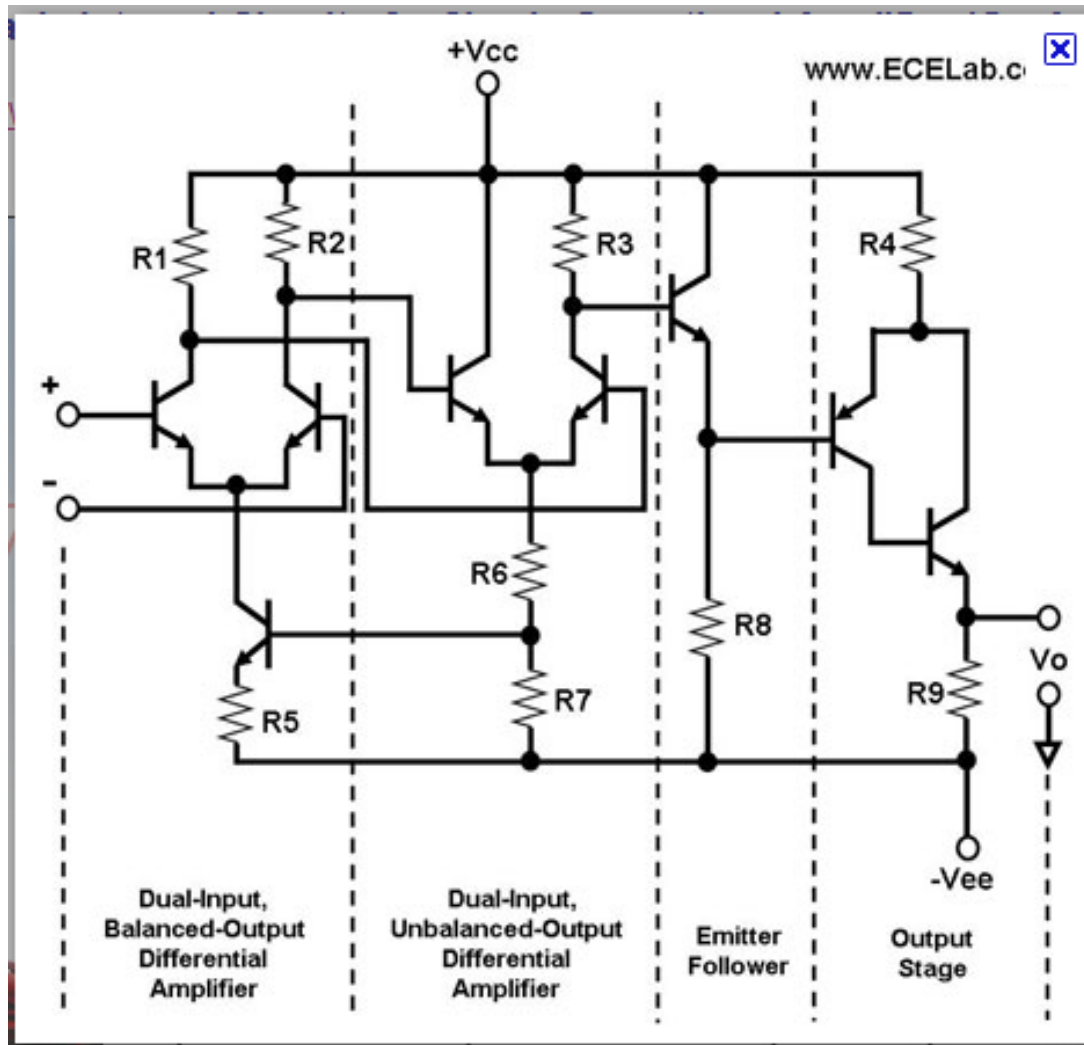
- Solve circuits with ideal op amps;
- Explain the effects of saturation, and check to see whether an op amp is saturated;
- Select op amp circuits and specify circuit parameters to meet design specifications;
- Draw, analyze, and explain the effects of non-ideal op amp models on circuit operation.

Original vacuum tube op-amp

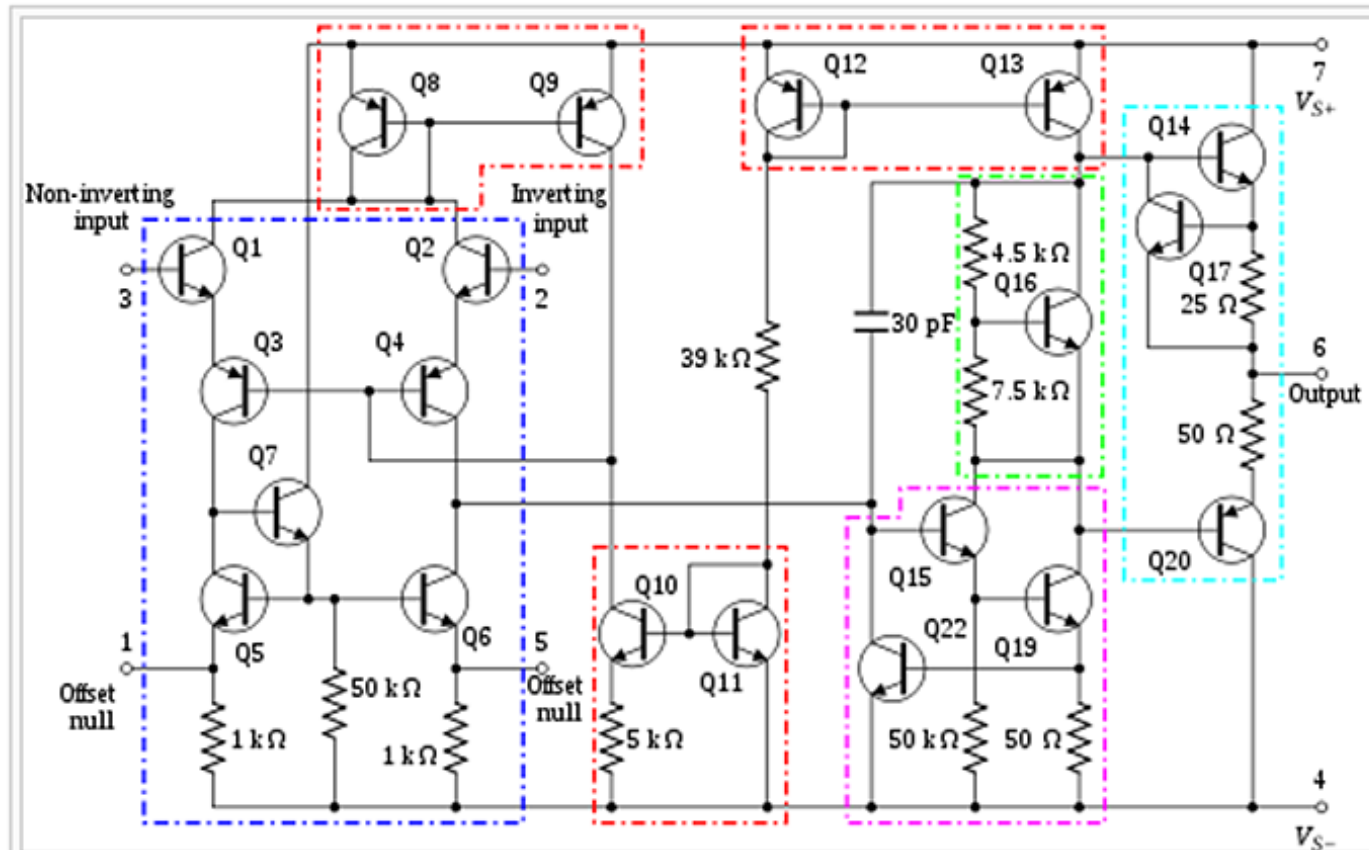
The Philbrick Research op-amp, model K2-W



What's inside an op-amp? (simplified view)



What's inside an op-amp? (actual view)

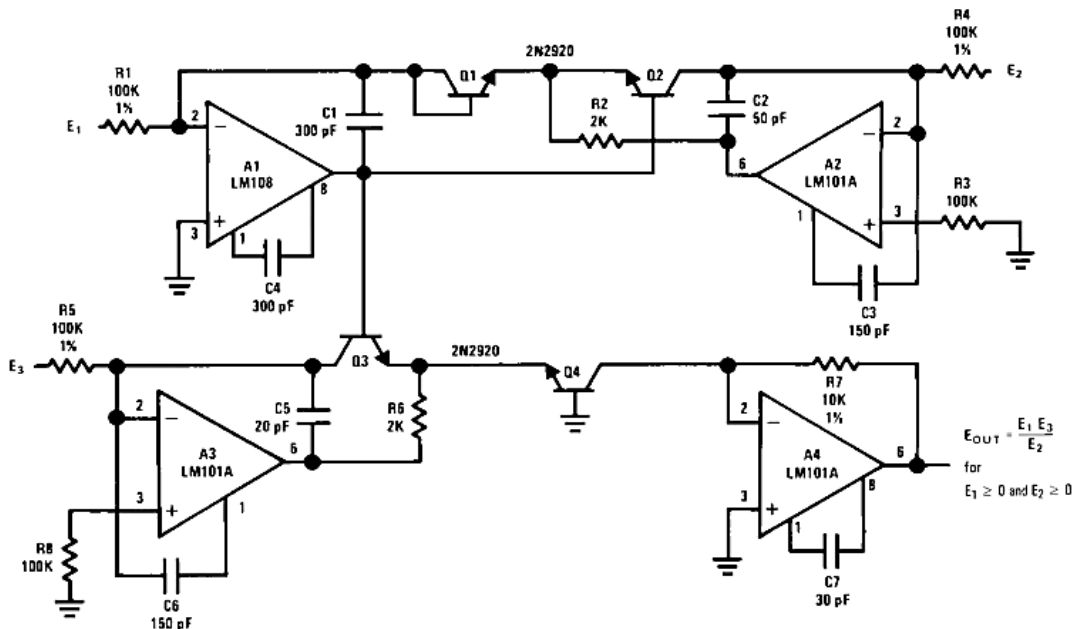


A component level diagram of the common 741 op-amp. Dotted lines outline: current mirrors (red); differential amplifier (blue); class A gain stage (magenta); voltage level shifter (green); output stage (cyan).

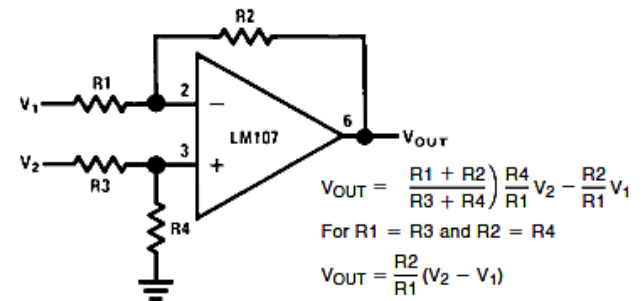
Why the op amp?

- The operation amplifier was so named because it could be used to perform mathematical operations such as:
 - Addition and subtraction
 - Multiplication and division
 - Log and antilog
 - Exponentiation

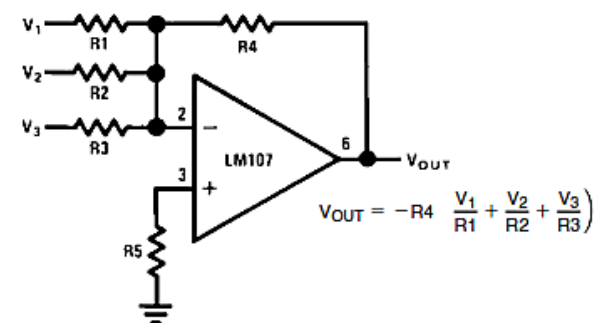
Multiplier Divider



Difference Amplifier



Inverting Summing Amplifier



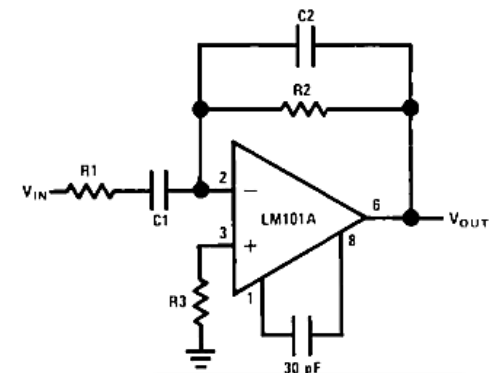
Analog computer

- The operation amplifier is the basic element of an analog computer
 - Performs integration and differentiation
- Can solve differential equations by simply connecting op amp blocks together with patch cords and adjusting the circuit gain to represent the constants in the equation



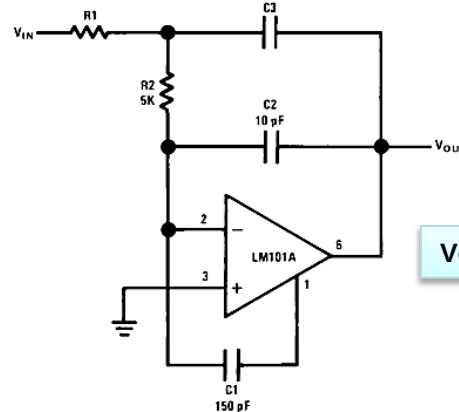
Analog Computer on a WWII Submarine

Practical Differentiator



$$V_{out} = K \cdot dV_{in}(t) / dt$$

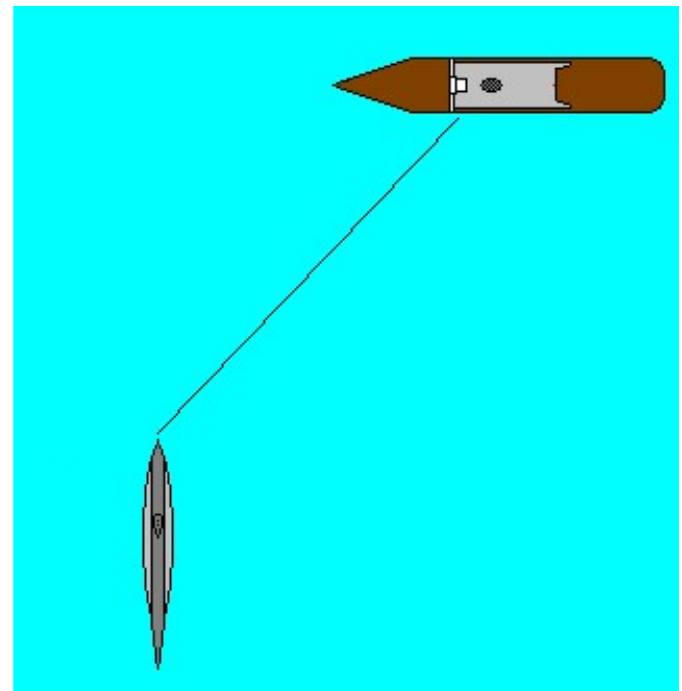
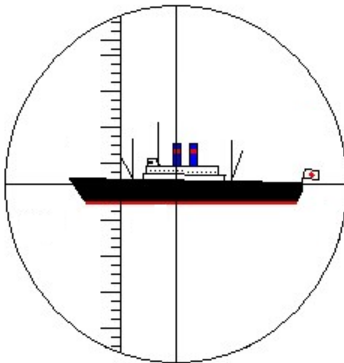
Fast Integrator



$$V_{out} = K \cdot \int V_{in}(t) dt$$

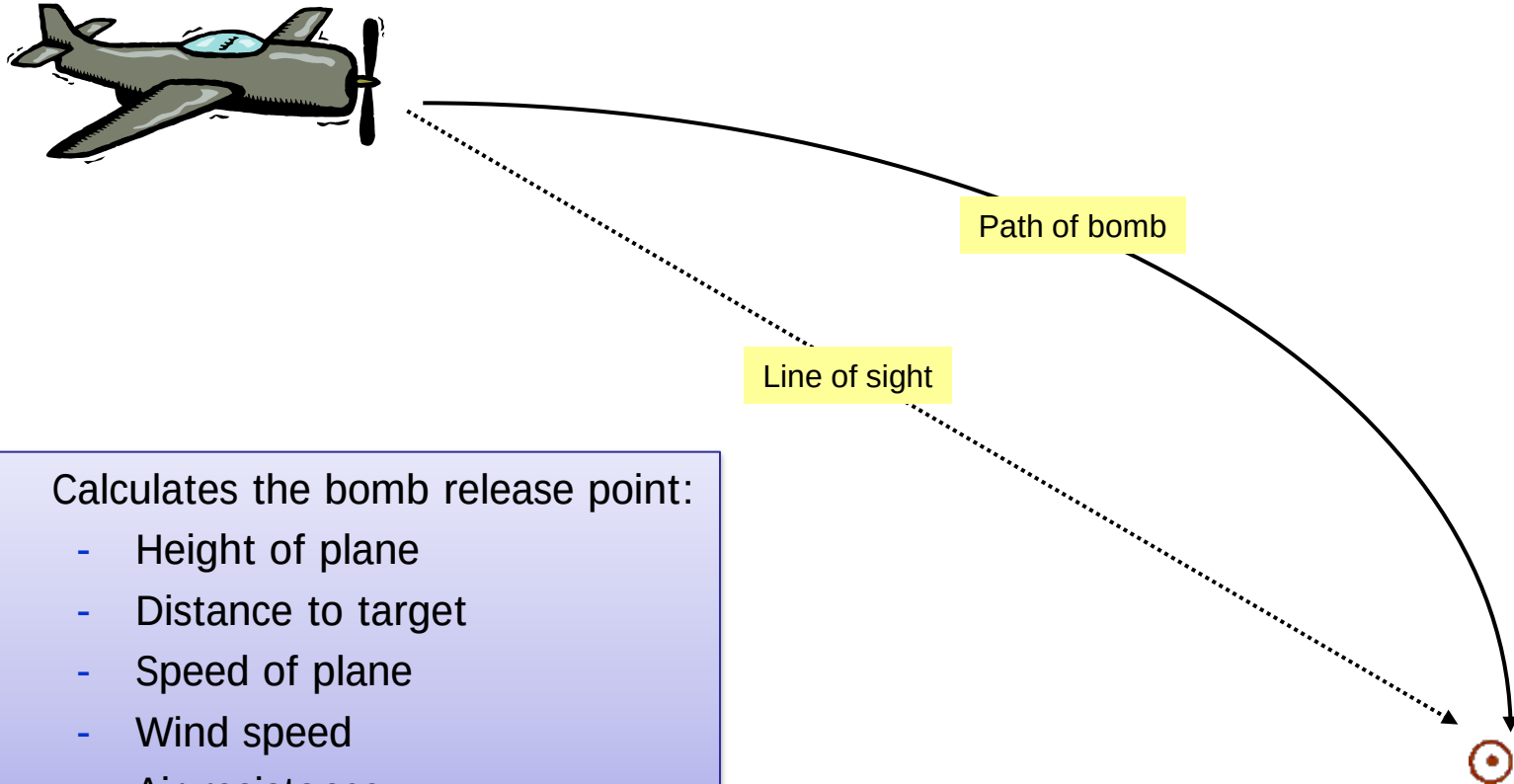
A problem in geometry

- Torpedo Data Computer (TDC) received information from the captain looking at the target through the periscope
- Input data:
 - Angle between the two ships
 - Speed of sub and target
 - Range to the target
- Output:
 - When to fire the torpedo
 - Intercept course heading for the torpedo



Dropping bombs

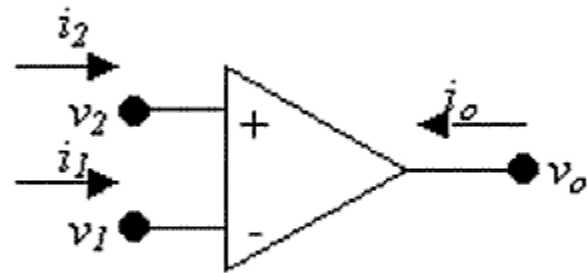
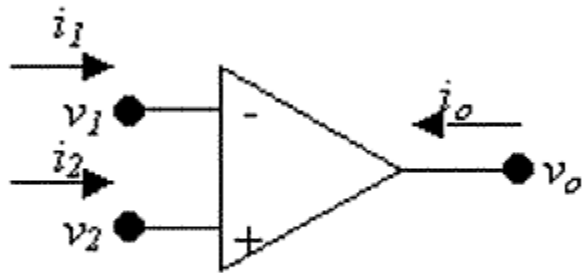
WW II bombers carried an analog computer, the Norden Bombsight



- Calculates the bomb release point:
 - Height of plane
 - Distance to target
 - Speed of plane
 - Wind speed
 - Air resistance

Circuit symbol

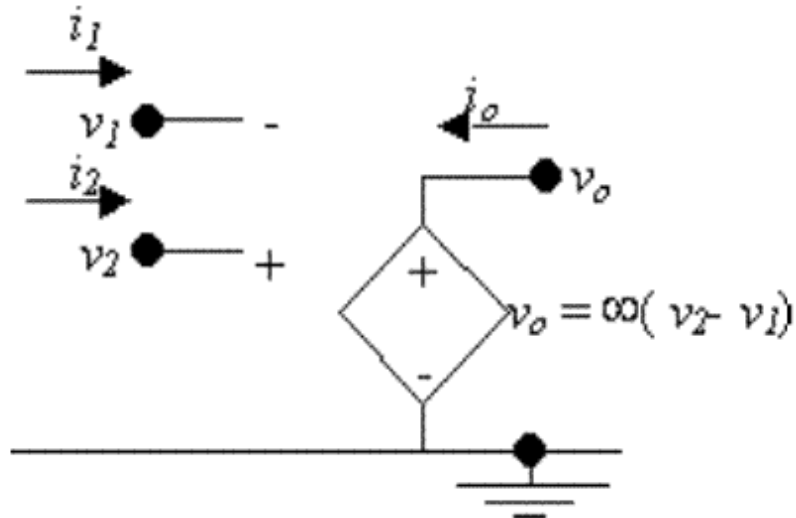
The op amp has a special circuit symbol, a triangle with two inputs and one output



The plus (+) and minus (-) connections are the non-inverting input and the inverting input. These are usually, but not always, drawn with the minus (-) on the top.

Ideal op amp relationships

An ideal op amp is an ideal voltage controlled voltage source. We can think of the op amp symbol being replaced as follows:



Notice what this implies: If v_o is finite, then

$$v_2 - v_1 = \frac{v_o}{\infty} = 0$$

$$v_1 = v_2$$

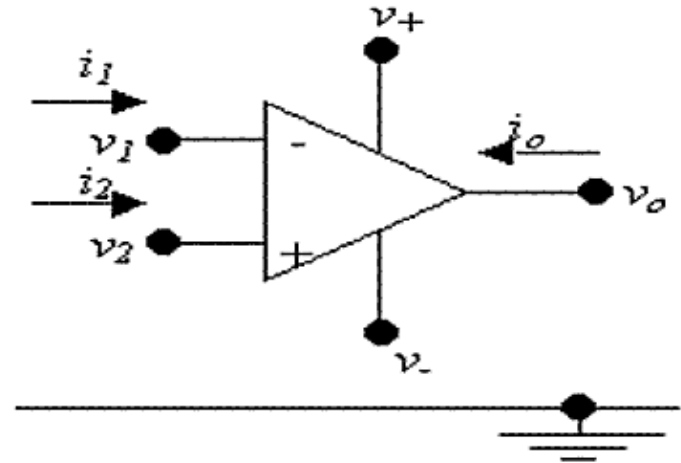
and

$$\begin{aligned} i_1 &= 0 \\ i_2 &= 0 \end{aligned}$$

These are, in fact, the *ideal op amp relationships*, and once we get used to them we can use them to solve op amp circuits.

Ideal op amp relationships-2

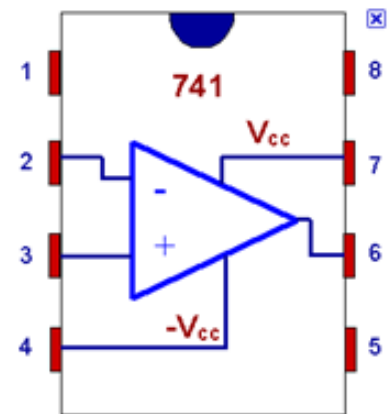
(At this point you might be wondering why the op amp has its own special symbol, when it's really just a VCVS. The answer has to do with packaging. The op amp comes in a single package that includes the inverting and non-inverting inputs. The op amp symbol indicates that the package combines all the connections in one place, where the VCVS symbol lets the inputs be anywhere.



op amp power supplies

But wait, if the input currents i_1 and i_2 are zero, where does the output current i_o come from? The dependent voltage source is used in the model, but in real life we have to supply the op amp with an external power supply. Many real world op amps (the LM741, for example) require both positive and negative voltage (as opposed to positive and ground, called single source).

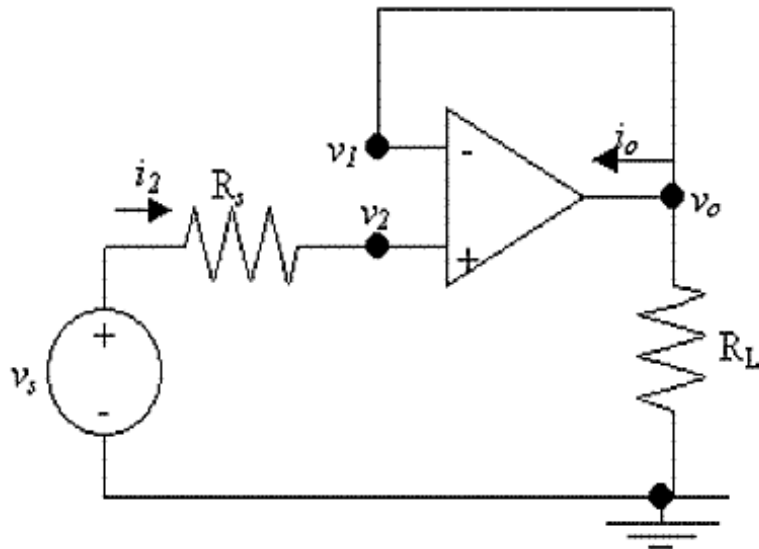
If we really want to do KCL for the op amp (for a supernode containing the op amp, or for the output node), we have to include the power supply. Usually, it is not necessary to include the power supply, and the power supply connections are omitted. We all understand that they are there, so why clutter up the circuit with them? And usually, it is not necessary to do KCL for a supernode containing the op amp or for the output node. We will do KCL for the input nodes.



8-pin Dual Inline Plastic (DIP) package

Op amp circuit analysis

Let's try a circuit with an op amp.
What's the relationship between v_s and v_o ?



$v_o = v_1$ since they are connected.

$v_o = v_1 = v_2$ from the ideal op amp relationships.

Remember that there is no current flowing into the op amp inputs. So

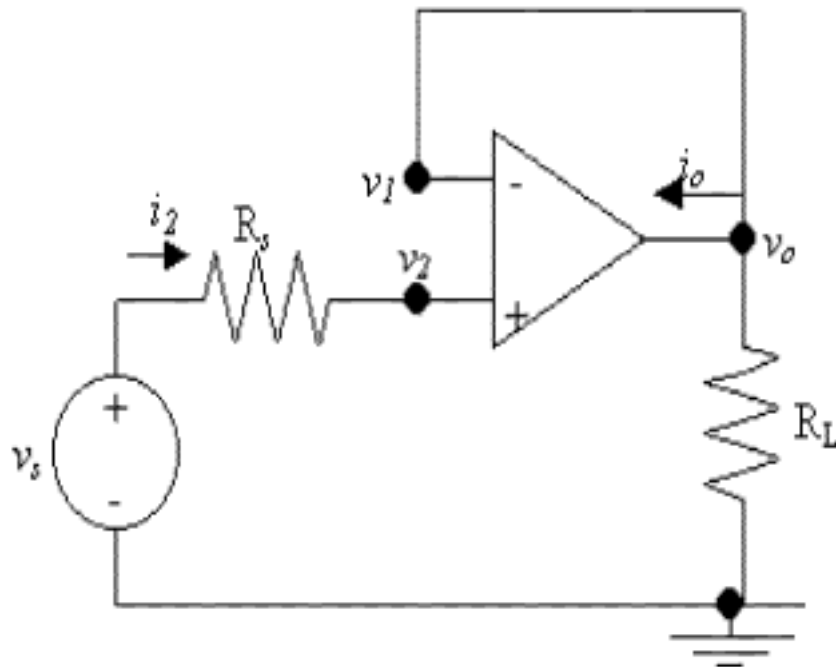
$i_2 = 0$, $v_2 = v_s$ and thus

$v_o = v_s$

See how the special op amp relationships simplify circuit analysis?

Op amp circuit analysis-2

This circuit is called a voltage follower, or buffer amplifier



At first glance, this circuit doesn't seem to do very much. The output voltage equals the input voltage. But consider the source (connected to in non-inverting input, v_2) before it is connected to the op amp.

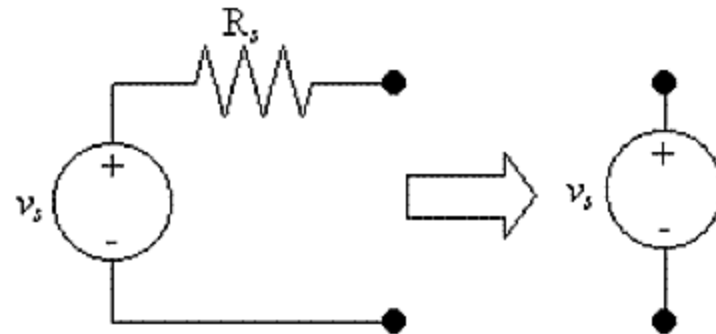
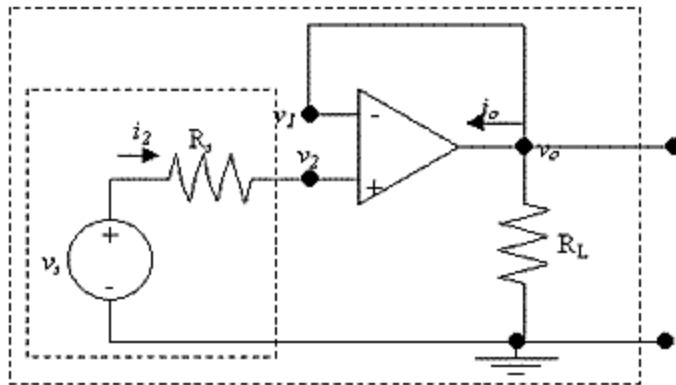
What is the Thévenin equivalent resistance looking into the source?

The source is already a Thévenin equivalent circuit, so: $R_t = R_s$

Op amp circuit analysis-3

Now, consider the Thévenin equivalent resistance seen looking in to the whole circuit.

v_{oc} is just v_o , and i_{sc} is - well, it's infinite! The op amp will hold $v_o = v_s$ no matter what the load resistance is.



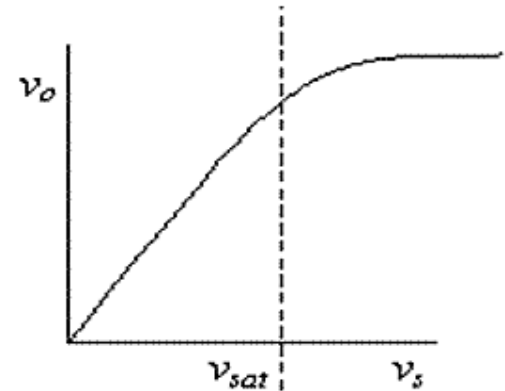
Because for ideal op-amp
 $i_2 = 0$ and hence $R_s = 0$,
there is no resistance
looking into the OC

the Thévenin equivalent is an ideal voltage source of value v_s :

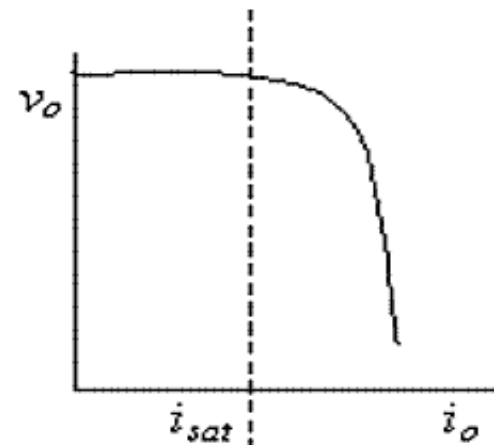
The voltage follower has eliminated the source impedance.

Saturation

If we actually built one of these voltage followers, and varied the input voltage v_s , we would see an interesting relationship between input and output. As the input voltage increased, output voltage would eventually start to lag behind, and finally not change at all.



If we kept v_s constant, and varied R_L so the output current varied, we would also see something unexpected. The output voltage would be constant for a while, and then, as current increased further, start to fall.



Saturation-2

What's happening is that the op amp is saturating, that is, moving out of the linear region and ceasing to operate as an amplifier. We can only model the op amp as ideal when it is not saturated.

We can check saturation, though, by comparing output voltage and current to saturation limits:

$$|v_o| \leq v_{sat}$$

$$|i_o| \leq i_{sat}$$

v_{sat} and i_{sat} come from the specification sheet. v_{sat} is usually a volt or two less than the power supply voltage.



Saturation-3

For example, if the power supply is $\pm 12\text{V}$ and i_{sat} is 2mA , then the maximum output voltage is 10V if V_{sat} is 10V .

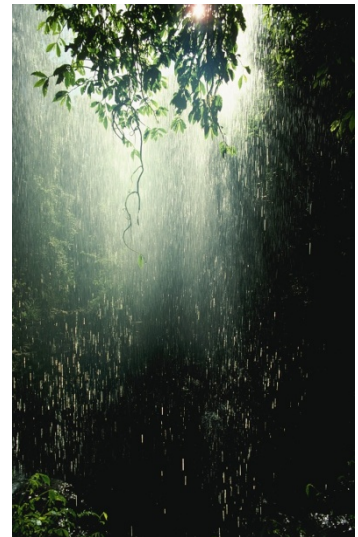
If V_{sat} is 10V , what's the minimum R_L ?

$$i_o \leq i_{sat}$$

$$\frac{v_o}{R_L} \leq 2\text{mA}$$

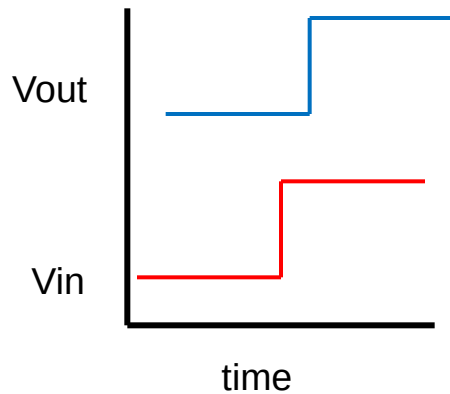
$$\frac{10}{R_L} \leq 2\text{mA}$$

$$R_L \geq \frac{10}{2 \times 10^{-3}} = 5\text{k}\Omega$$

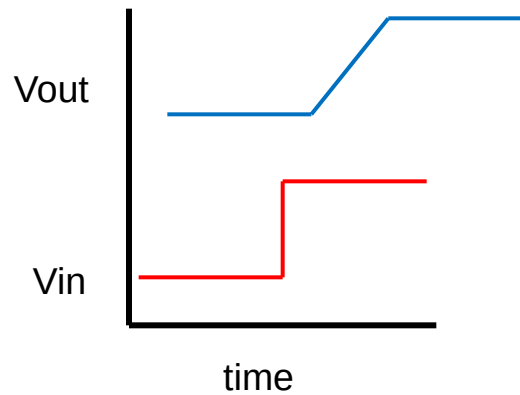


Slew rate

At this point, we should probably note that another departure from ideality in the op amp is that the slew rate, $\frac{dv_o}{dt}$, is also limited. This affects how fast the op amp responds to a step change in input, and how high a frequency it can operate at. We won't pay much attention to slew rate in this course, but you will run into it in later courses, and in the real world!



Ideal op amp

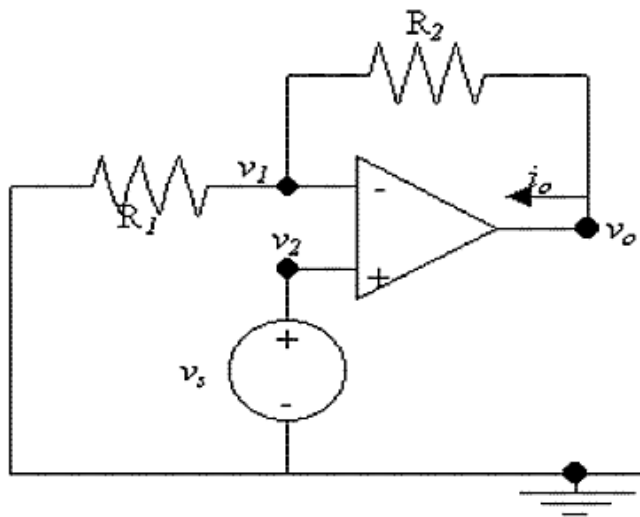


Real op amp

The slew rate for the 741 op amp is 0.5V per microsecond

More analysis

Let's look at another circuit (you should try the analysis)



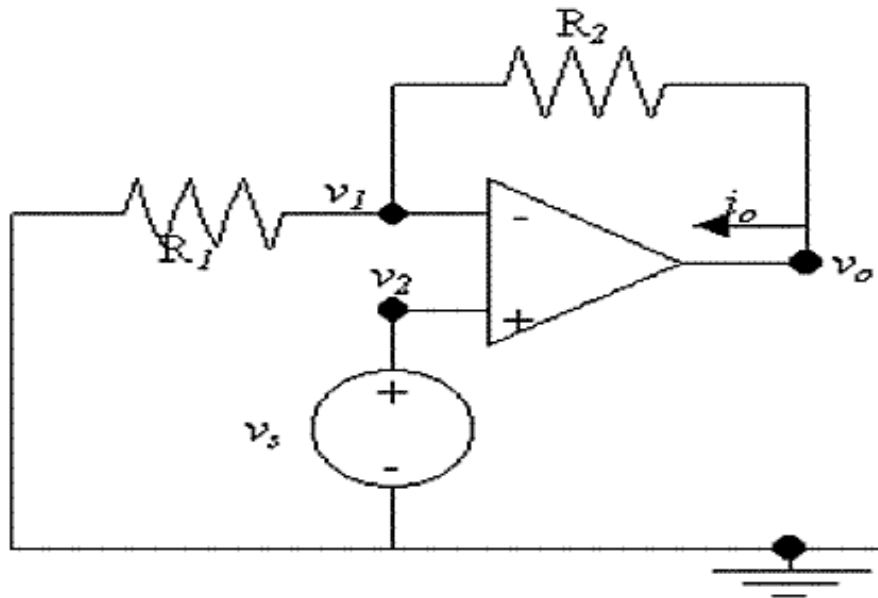
As before, find v_o as a function of v_{os} .

First, what's v_2 ? v_s .

So, what's v_1 ? v_s

This is a good time to practice active learning. Stop here and try to solve for v_o as a function of V_{os} . You can always advance to the next slide and see the answer, but that's no fun!

More analysis-2



Now, write KCL at v_1 :

$$\frac{v_s}{R_1} + \frac{v_s - v_o}{R_2} = 0$$

$$v_s \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{v_o}{R_2} = 0$$

$$v_o = v_s \left(1 + \frac{R_2}{R_1} \right)$$

Key term

- Potentiometer

A three-terminal resistor with a center sliding contact (wiper) that forms a voltage divider or a variable resistor



This is called a **non-inverting amplifier**. Notice that the gain of the amplifier can be changed by changing the ratio of the resistors, for example with a potentiometer for R_2 .

Design with op amps

Somewhere, in your textbook* (or some textbook somewhere!), is a catalog of op amp circuits that perform different functions. There are lots, beyond what's in the book. The circuits are generic; that is, they do not have specific values for the resistances. The catalog is a starting point for circuit design. We now know enough to do some almost real-world design. The process is:

- Given a function to perform
- Choose a circuit from the catalog that performs the function.
(Or, invent a new one. If you do, patent it.)
- Figure out values for the circuit components so that the specific function we want is implemented (e.g., specific gain).
- Check the circuit against limits.
- Iterate as required.

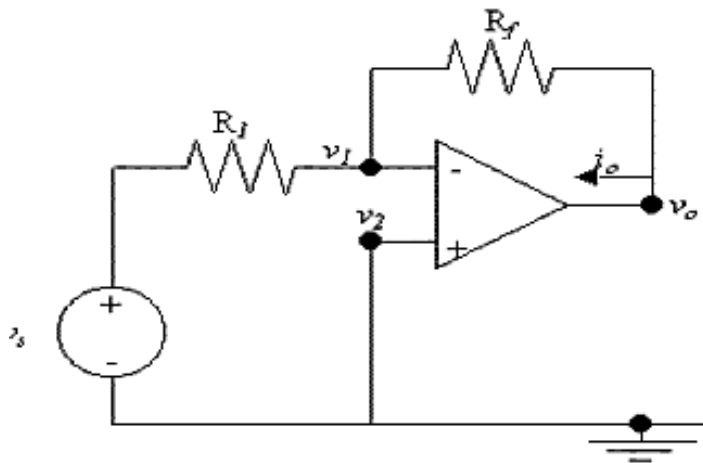
* www.stanford.edu/class/ee122/.../op_amp_circuit%20collection_AN-31.pdf

Design with op amps-2

Example: We want to amplify an input signal by a factor of 100 (gain of 100). The input signal has a range of 0 to -100 mV, but we want a positive voltage output.

Pretty clearly we want the inverting amplifier circuit from the catalog:

Please wait while I thumb through the catalog. OK, I found it:



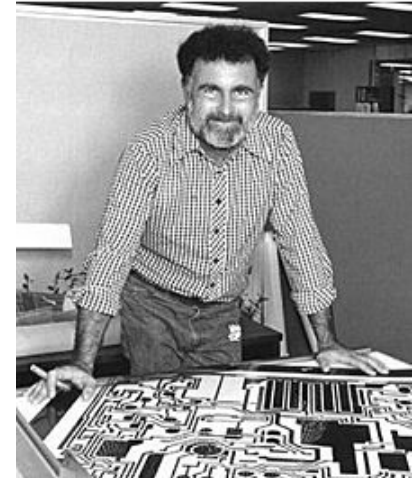
Let's check the analysis:

$$v_2 = 0$$

$$v_1 = 0$$

$$\frac{v_s - 0}{R_1} + \frac{v_o - 0}{R_f} = 0$$

$$v_o = -\frac{R_f}{R_1} v_s$$



Robert Widlar inventor of the first mass-produced op amp, uA709